

Question 1.

1.1 If (x^*, y^*, z^*) is a solution of the system of linear equations given by

$$2x - 3y + z = 8,$$

$$-x + 4y + 2z = -4,$$

$$3x - y + 2z = 9,$$

then find the value of $x^* + y^* + z^*$.

(6 points)

วิทยาลัยเทคโนโลยีอุตสาหกรรม

1.2 Solve the system of nonlinear equations as follows

$$\frac{1}{x^2} + \frac{1}{y^2} = 13,$$

$$\frac{1}{x} + \frac{1}{y} = 5.$$

(6 points)

วิทยาลัยเทคโนโลยีอุตสาหกรรม

Question 2.

2.1 Find all positive integer values of a such that $(a - 2) \mid 18$.

(4 points)

2.2 Assume a is a nonzero integer and let p and q be integers. If $a \mid (3p + 9q)$ and $a \mid (2p + 4q)$, then show that $a \mid 30q$.

(6 points)

วิทยาลัยเทคโนโลยีอุตสาหกรรม

Question 3. Let G be the greatest common divisor (gcd) of 940 and 4260.

3.1 Use the Euclidean algorithm to find the value of G . (4 points)

3.2 Given that $G = 4260x + 940y$, find the value of $x + y$. (4 points)

3.3 Find the least common multiple (lcm) of 940 and 4260. (2 points)

วิทยาลัยเทคโนโลยีอุตสาหกรรม

Question 4. Find the square roots of $29 - 4\sqrt{51}$ by answering in the simplest form.
(4 points)

วิทยาลัยเทคโนโลยีอุตสาหกรรม

Question 5. Determine whether the following equation

$$\sqrt{x-2} = \sqrt{2x+3} + 2$$

has any solution(s). If so, find the solution(s). If not, explain your reasoning. (5 points)

วิทยาลัยเทคโนโลยีอุตสาหกรรม

Question 6.

6.1 Solve the equation $4^x - 7(2^x) - 8 = 0$.

(4 points)

6.2 Solve the inequality $\left(\frac{3}{4}\right)^{x^2+2x+8} \leq \left(\frac{3}{4}\right)^{2x+24}$.

(5 points)

วิทยาลัยเทคโนโลยีอุตสาหกรรม

Question 7.

7.1 Evaluate the value of $\log_2 16 + \ln e^2 - 3^{\log_3 4}$.

(4 points)

7.2 Given that x, y and z are positive numbers with $y \neq 1$ and $z \neq 1$.

Solve the equation $(\log_y x)(\log_z y)(\log_{25} z) = \log_5 125$ for x . (4 points)

วิทยาลัยเทคโนโลยีอุตสาหกรรม

7.3 Solve the logarithmic equation $\log_4 \log_3 \log_2 (3x^2 - 4) = 0$.

(4 points)

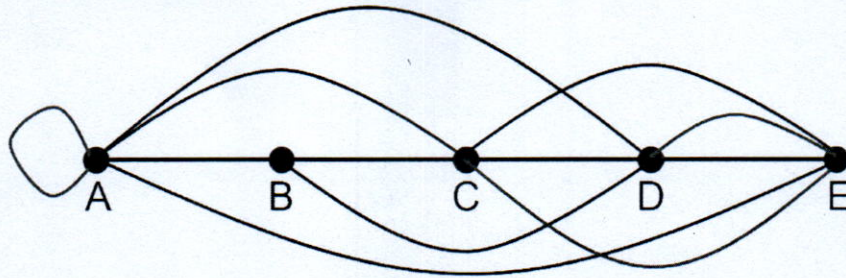
7.4 Solve the logarithmic inequality $\log_3 (x + 3) > \log_3 (2x + 1)$.

(4 points)

วิทยาลัยเทคโนโลยีอุตสาหกรรม

Question 8. Given a graph $G = (V, E)$ as follows.

(6 points)

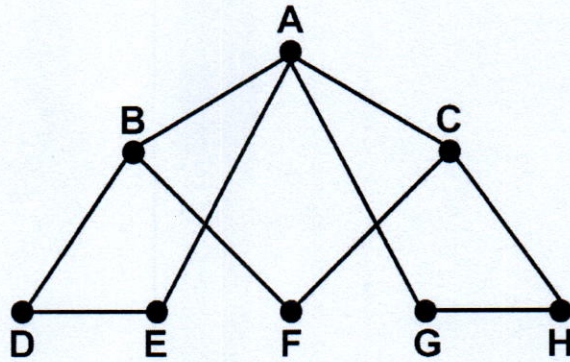


- Determine whether the graph is a directed or undirected graph.
- Determine whether the graph is a weighted or unweighted graph.
- Determine the set of vertices $V(G)$.
- Determine the number of edges in this graph.
- Determine the degree of node E ($\deg(E)$).
- Determine the highest degree of a node in this graph.

วิทยาลัยเทคโนโลยีอุตสาหกรรม

Question 9.

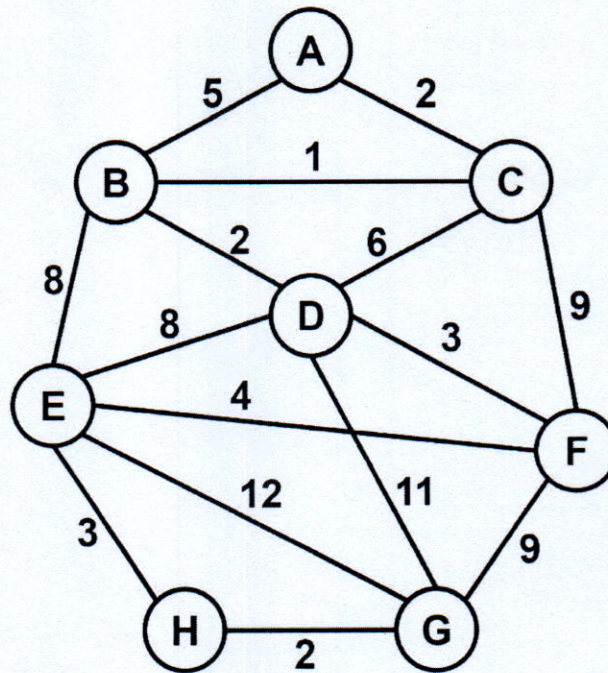
9.1 Consider the following graph:



Examine whether the given graph has an Euler circuit. If one exists, provide the Euler circuit for the graph. If the graph does not have an Euler circuit, identify the edge(s) that need to be added to enable the creation of an Euler circuit. (4 points)

วิทยาลัยเทคโนโลยีอุตสาหกรรม

9.2 Find the shortest path between the vertex A and the vertex G in the following weighted graph. (6 points)



วิทยาลัยเทคโนโลยีอุตสาหกรรม